Analysis and Design of Structural Sandwich Panels Against Denting

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Abstract: The problem of local bending often encountered in laterally loaded sandwich panels is analyzed by considering the deflection of the loaded facing against the not loaded facing as being governed by an elastic foundation model. This is achieved by the development of a two-parameter elastic foundation formulation which takes into account the existence of shear interactions between the loaded facing and the supporting medium (core material). The results obtained by application of the developed approximate solution procedure, i.e., the local stress field in the near vicinity of the area of application of the external load, are compared with finite element solution results and very good comparative results are obtained.

Keywords: structural sandwich panels / indentation / local bending effects / elastic foundation / classical sandwich theory / comparative study / finite element analysis / parametric effects / design tool / concurrent engineering.

1 Introduction

The structural member, known as a sandwich panel, is a special form of a laminated composite which consists of three principal parts; thin, strong, and relatively high density facings separated by a thick, light, and weaker core. The faces and the core of the sandwich panel are adhesively joined in order to transfer the load between the components. This provides a structure which is very efficient for resisting bending and buckling loads since the sandwich panel is much stronger and stiffer, in most respects, than the sum of the individual strengths and stiffnesses.

Although sandwich structures have been accepted as an excellent way to design strong, durable and lightweight structures, a number of important problems have been left more or less unattended, even by classical textbooks as the ones by Allen [1], Plantema [2] and Stamm & Witte [3]. An area of significant practical importance, and belonging to this class of more or less unattended problems, is the so called denting problem associated with local

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bending effects in structural sandwich panels. These local bending effects appear more frequently than is usually expected, and two typical examples of areas/positions where local bending can be observed are: 1. various joints between sandwich panels (T-joints, corner-joints); 2. inserts and fasteners for introduction or "take-out" of external forces as well as for mechanical fastening of various types of equipment.

It is a well known fact that sandwich panels are notoriously sensitive to failure by the application of strongly localized external lateral loads, i.e., point loads, line loads and distributed loads of high intensity. This pronounced sensitivity toward the application of concentrated external loads is due to the associated inducement of significant local deflections of the facing into the core material. The result of this may be a premature failure of the sandwich panel.

The present paper is exclusively devoted to local bending analysis of sandwich panels (primarily foam-cored) which can fail in several distinctly different ways [1-5]. Possible failure modes in the present context could be crushing (denting failure) of the core material [4-7]; delamination at the interface between the loaded face and the core material due to the concentration of interlaminar stresses in the region adjacent to the area of external load application; shear failure of the core material; tensile or compressive failure of the facings by yielding or fracturing dependent on the properties of the face-materials; and finally delamination due to interlaminar shear and transverse normal stresses for the case of laminated FRP-faces.

Relatively few references have been treating the problems associated with local bending of the faces of sandwich panels subjected to concentrated loads, and no one gives an explicit description of the onset and development of actual failure modes. They restrict themselves to the development of structural analyses based on the somewhat unrealistic assumption of linear elastic behavior of the core material as well as the faces of the considered sandwich panel. Thus, the results obtained do not reflect the actual sequence of events experienced by the constituent materials during the onset and development of failure, but they do have the potential of giving valuable information about the fundamental mechanics of the local bending problem as well as the parameters controlling the onset of irreversible failure.

In the modelling of the problem, it seems reasonable to consider the relative deflection of the loaded face of the sandwich panel against the not loaded face as being governed by an appropriate elastic foundation formulation.

This approach was used by Reference 6, who considered the foam-core as continuously distributed linear tension/compression springs supporting the loaded face of the sandwich panel. Thus, the elastic response of the foam-core material was assumed to be governed by the "classical" Winkler foundation model, which is extensively treated by, among others, Hetényi [8].

However, the Winkler foundation model suffers a serious drawback, since it does not account for the possible existence of shear interactions between the loaded face and the supporting medium (core material). This feature of the Winkler foundation model suggests that it becomes inadequate for deformations of short wave-length, in which the shear deformations of the foam-core of the sandwich beam become important.

2 Elastic Foundation Analogy

2.1 Formulation of Two-Parameter Foundation Model

The present formulation is based on the assumption of an elastic foundation model, which does take into account the existence of shear interactions between the loaded face and the supporting medium (core material). This additional consideration, compared to the Winkler foundation model, is necessary in order to account for the build-up of interfacial shear stresses occurring adjacent to the local area of external load application.

Figure 1 illustrates the considered problem with its constituent parts: the loaded facing of a sandwich beam and the supporting medium (core material). The considered facing can be subjected to arbitrary distributed external transverse normal and shear loads $p_Z(x)$, $p_X(x)$ along its upper surface.

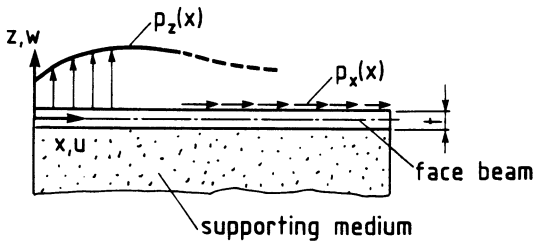


Figure 1. Loaded Facing Supported by Core Material.

It is assumed that the face as well as the core material can be satisfactory described by use of linear elastic constitutive relations, even though this assumption in general has to be considered as being unrealistic as mentioned earlier.

The elastic response of the supporting medium (core material) is suggested expressed by the following equations, which relates the deflections of the loaded facing to the interfacial stress components measured per unit length of the face-beam (Figure 1):

$$q_x(x) = K_x\, u\left(x,-\frac{t}{2}\right)\ ;$$
$$q_z(x) = K_z\, w(x)\ ; \tag{1}$$

where K_x, K_z : foundation moduli;
$q_x(x)$, $q_z(x)$: interfacial shear and normal stress resultants;
$u(x,-t/2)$: longitudinal displacement of lower fibre of the facing;
$w(x)$: lateral displacement of the loaded facing;
x : independent variable (longitudinal coordinate).

It should be emphasized that $q_x(x)$, $q_z(x)$ represents the foundation normal and shear stress resultants per unit length of the beam (unit: N/mm), i.e., they can be obtained by multiplying the stress components $\tau_{zx}(x)$, $\sigma_z(x)$ at the interface with the width of the beam (2b). In the forthcoming derivations $q_x(x)$, $q_z(x)$ will be referred to as the interlaminar stress distribution functions.

The foam-core and the faces are assumed to be homogeneous, isotropic and linear elastic, and the foundation modulus K_z is suggested related to the elastic coefficients of the core (E_c,ν_c) and the loaded facing (E_f) by the following expression [8][1]:

$$K_z = 0.71\, E_c \sqrt[3]{\frac{E_c\, b^4}{E_f\, I}}\ ; \tag{2}$$

where b : half width of the face-beam;
E_f : elastic modulus of face-beam material;
I : area moment of inertia of the face-beam;
$E_f I$: flexural rigidity of face-beam.

Equation 2 is developed under the assumption that the height t (see Figure 1) of the face is small compared to the wave-length of the applied load [8].

The second foundation modulus K_x is suggested related to K_z through the relation:

$$K_x = \frac{K_z}{2(1+\nu_c)}\ . \tag{3}$$

The loaded facing is modelled by application of the classical theory of bending of beams. Thus, it is assumed that normals to the undeformed neutral axis of the face-beam remains straight, normal and inextensional during deformation so that transverse normal and shearing strains may be neglected in deriving the beam kinematic relations. The displacement

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[1] Obtained by comparison with 2-D elasticity solution to the problem of an infinitely long beam attached to an elastic medium which extends to infinity on one side.

components u, w of an arbitrary point in the face-beam may be expressed by the neutral axis quantities u_0, w_0 by the equations:

$$u = u_0 - z\frac{dw}{dx}; \qquad w = w_0 \tag{4}$$

The first of Equations 1 together with Equation 4 yields:

$$q_x(x) = K_x\left(u_0 + \frac{t}{2}\frac{dw}{dx}\right). \tag{5}$$

From Equation 5 it is seen that the interfacial shear stress distribution function $q_x(x)$ is proportional to the angular rotation dw/dx of the neutral axis of the face-beam. The interfacial stress distribution functions $q_z(x)$, $q_x(x)$ are unknown, and the objective of the proceeding derivations is to formulate and solve the equations determining these functions.

Referring to Figure 2 for notation and sign convention, the three equilibrium equations for the elastically supported face-beam can be written in the form:

$$\frac{dN}{dx} = q_x - p_x; \qquad \frac{dQ}{dx} = q_z - p_z; \qquad \frac{dM}{dx} = Q - \frac{t}{2}(q_x + p_x). \tag{6}$$

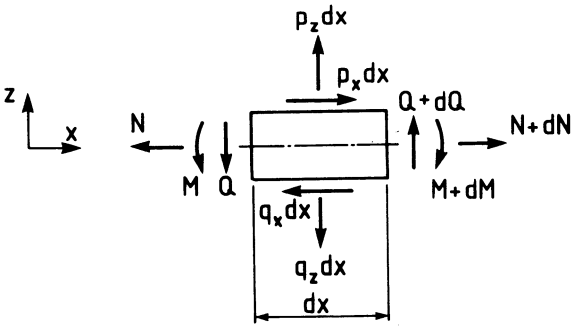


Figure 2. Notation and Sign Convention for Equilibrium of Loaded Face-Beam.

The longitudinal normal strain ε_x at the interface between the facing and the core material is expressed as follows, where a unit width (2b=1) sandwich beam is considered:

$$\varepsilon_x\left(x,-\frac{t}{2}\right)=\frac{d}{dx}\left(u\left(x,-\frac{t}{2}\right)\right)=\frac{1}{E_f}\,\sigma_x\left(x,-\frac{t}{2}\right)\Rightarrow$$

$$\frac{d}{dx}\left(u\left(x,-\frac{t}{2}\right)\right)=\frac{1}{E_f t}\left(N-\frac{6M}{t}\right). \tag{7}$$

From the first of Equation 1 the following is derived:

$$\frac{dq_x(x)}{dx}=K_x\frac{d}{dx}\left(u\left(x,-\frac{t}{2}\right)\right). \tag{8}$$

By combination of Equations 7 and 8 followed by successive differentiation as well as introduction of the equilibrium Equation 6, the following expressions appear:

$$\frac{dq_x}{dx}=\frac{K_x}{E_f t}\left(N-\frac{6M}{t}\right)\Rightarrow$$

$$\frac{d^2q_x}{dx^2}=\frac{K_x}{E_f t}\left(4\,q_x-\frac{6}{t}\,Q+2\,p_x\right)\Rightarrow \tag{9}$$

$$\frac{d^3q_x}{dx^3}=\frac{K_x}{E_f t}\left(4\,\frac{dq_x}{dx}+2\,\frac{dp_x}{dx}-\frac{6}{t}\,(q_z-p_z)\right).$$

By rearrangement, the latter of Equation 9 can be written in the form:

$$\frac{d^3q_x}{dx^3}-\frac{4\,K_x}{E_f t}\,\frac{dq_x}{dx}=-\frac{6\,K_x}{E_f t^2}\,(q_z-p_z)+\frac{2\,K_x}{E_f t}\,\frac{dp_x}{dx}\,. \tag{10}$$

Equation 10 establishes the relationship between the interfacial stress distribution functions $q_x(x)$, $q_z(x)$.

A similar equation can be found by combining the latter of Equation 1 with the "flexure formula" known from the classical theory of bending of beams:

$$q_z=K_z\,W\;\Rightarrow$$

$$\frac{d^2q_z}{dx^2}=K_z\,\frac{d^2w}{dx^2}=-\frac{K_z}{E_f I}\,M\;\Rightarrow$$

$$\frac{d^3q_z}{dx^3}=-\frac{K_z}{E_f I}\left(Q-\frac{t}{2}\,(q_x+p_x)\right)\Rightarrow \tag{11}$$

$$\frac{d^4q_z}{dx^4}=-\frac{K_z}{E_f I}\left(q_z-p_z-\frac{t}{2}\left(\frac{dq_x}{dx}+\frac{dp_x}{dx}\right)\right).$$

Rearrangement of the latter of Equation 11 gives:

$$\frac{d^4 q_z}{dx^4} + \frac{K_z}{E_f I} q_z = \frac{t}{2} \frac{K_z}{E_f I} \left(\frac{dq_x}{dx} + \frac{dp_x}{dx} \right) + \frac{K_z}{E_f I} p_z \,. \tag{12}$$

Equations 10 and 12 represent a set of two coupled ordinary constant-coefficient differential equations expressed in the two unknown interfacial stress distribution functions.

In order to obtain two differential equations expressed solely in the two unknown functions $q_x(x)$ and $q_z(x)$, Equation 10 can be employed to eliminate $q_z(x)$ and its derivatives from Equation 12. Similarly Equation 12 can be employed to eliminate the derivatives of $q_x(x)$ from Equation 10.

From Equation 10 the following relations are obtained:

$$q_z = -\frac{E_f t^2}{6 K_x} \left(\frac{d^3 q_x}{dx^3} - \frac{4 K_x}{E_f t} \frac{dq_x}{dx} \right) + \frac{t}{3} \frac{dp_x}{dx} p_z \;\Rightarrow$$

$$\frac{d^4 q_z}{dx^4} = -\frac{E_f t^2}{6 K_x} \left(\frac{d^7 q_x}{dx^7} - \frac{4 K_x}{E_f t} \frac{d^5 q_x}{dx^5} \right) + \frac{t}{3} \frac{d^5 p_x}{dx^5} + \frac{d^4 p_z}{dx^4} \,. \tag{13}$$

Similarly Equation 12 gives:

$$\frac{dq_x}{dx} = \frac{2 E_f I}{K_z t} \left(\frac{d^4 q_z}{dx^4} + \frac{K_z}{E_f I} q_z \right) - \frac{2}{t} p_z - \frac{dp_x}{dx} \;\Rightarrow$$

$$\frac{d^3 q_x}{dx^3} = \frac{2 E_f I}{K_z t} \left(\frac{d^6 q_z}{dx^6} + \frac{K_z}{E_f I} \frac{d^2 q_z}{dx^2} \right) - \frac{2}{t} \frac{d^2 p_z}{dx^2} - \frac{d^3 p_z}{dx^3} \,. \tag{14}$$

Inserting Equations 13 and 14 into Equations 10 and 12, yields a set of two ordinary non-homogeneous differential equations of seventh and sixth order respectively, where each of the two differential equations is expressed solely in one of the unknown interfacial stress distribution functions $q_x(x)$, $q_z(x)$:

$$\frac{d^7 q_x}{dx^7} + \alpha_2 \frac{d^5 q_x}{dx^5} + \alpha_1 \frac{d^3 q_x}{dx^3} + \alpha_0 \frac{dq_x}{dx} = -\frac{3 \alpha_2}{2 t} \frac{d^4 p_z}{dx^4} - \frac{\alpha_2}{2} \left(\frac{d^5 p_x}{dx^5} - \frac{\alpha_1}{2} \frac{dp_x}{dx} \right); \tag{15}$$

$$\frac{d^6 q_z}{dx^6} + \alpha_2 \frac{d^4 q_z}{dx^4} + \alpha_1 \frac{d^2 q_z}{dx^2} + \alpha_0 q_z = \alpha_1 \left(\frac{d^2 p_z}{dx^2} + \frac{\alpha_2}{4} p_z \right) + \frac{\alpha_1 t}{2} \left(\frac{d^3 p_x}{dx^3} + \frac{\alpha_2}{2} \frac{dp_x}{dx} \right); \tag{16}$$

$$\alpha_2 = -\frac{4\,K_x}{E_f\,t}\;;\;\alpha_1 = \frac{K_z}{E_f\,I}\;;\;\alpha_0 = \frac{\alpha_1\,\alpha_2}{4}\;. \tag{17}$$

The complete solution to the governing set of two ordinary non-homogeneous constant-coefficient differential, Equations 15 and 16 can be written in the general form:

$$\begin{aligned} q_x(x) &= (q_x(x))_h + (q_x(x))_p\;; \\ q_z(x) &= (q_z(x))_h + (q_z(x))_p\;; \end{aligned} \tag{18}$$

where subscript "h" denotes the solutions to the homogeneous parts of Equations 15 and 16 and subscript "p" denotes the particular solutions to Equations 15 and 16.

The solution of the homogeneous part of Equations 15 and Equation 16 is straightforward, and it can be shown [9] that the homogeneous solution to Equation 15 can be written in the form:

$$\begin{aligned} (q_x(x))_h = A_0 &+ A_1\cosh(\phi_1 x) + A_2\sinh(\phi_1 x) \\ &+ A_3\cosh(\xi x)\cos(\eta x) + A_4\sinh(\xi x)\cos(\eta x) \\ &+ A_5\sinh(\xi x)\sin(\eta x) \;\; + A_6\cosh(\xi x)\sin(\eta x)\;; \end{aligned} \tag{19}$$

where ϕ_1, ξ, η are coefficients defined in terms of $\alpha_2, \alpha_1, \alpha_0$ given by Equation 17. ϕ_1, ξ, η are given in explicit form in Reference 9 to which the reader is referred for further details. A_j (j=0,1,...,6) are seven integrational constants which have to be determined from the statement of the boundary conditions of the problem.

From the first of Equation 13 (omitting the non-homogeneous terms) it follows, that the homogeneous part of the transverse normal stress response $(q_z(x))_h$ can be expressed by the first and third derivatives of $(q_x(x))_h$:

$$(q_z(x))_h = \frac{2\,t}{3\,\alpha_2}\left(\frac{d^3(q_x(x))_h}{dx^3} + \alpha_2\,\frac{d(q_x(x))_h}{dx}\right)\;. \tag{20}$$

Thus, the general solution to the homogeneous parts of the governing differential Equations 15 and 16 has been established.

The particular solution parts of Equations 15 and 16 still need to be determined, and this is accomplished by assuming suitable particular solution functions by "qualified quessing". The derivation of the particular solutions $(q_x(x))_p$, $(q_z(x))_p$ is of course strongly influenced by the form of the external loads, since these load distribution functions $p_x(x)$, $p_z(x)$ determine the non-homogeneous terms appearing in the governing differential equations.

For further information about the explicit expressions for the particular solution parts, for various types of external loading, the reader is referred to Thomsen [10,11].

2.2 Specification of Boundary Conditions

In order to determine the seven integrational constants A_j (j=0,1,...,6), it is necessary to formulate seven boundary conditions. To exemplify the formulation of the necessary boundary conditions the simplest possible example is considered: the loaded facing of a simply supported sandwich beam in 3-point bending (shown in Figure 3). The necessary boundary conditions for the loaded facing can be expressed as follows:

$$\left[q_x\right]_{x=0} = 0\,; \qquad \left[\frac{dw}{dx}\right]_{x=0} = 0\,;$$

$$\left[\frac{d^2w}{dx^2}\right]_{x=L} = 0\,; \qquad \left[\frac{d^3w}{dx^3}\right]_{x=L} = 0\,; \tag{21}$$

$$\int_0^L q_x(x)\,dx = 0\,; \qquad \int_0^L q_z(x)\,dx = \frac{P}{2}\,.$$

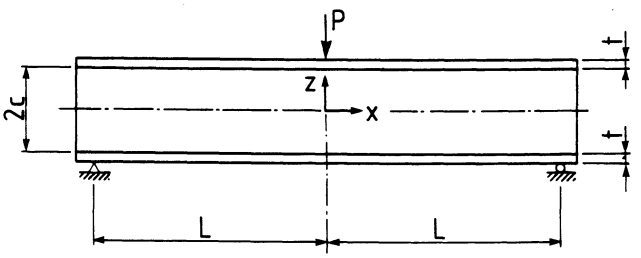


Figure 3. Unit Width Sandwich Beam in 3-Point Bending.

The first and the second of Equation 21 represent the conditions of symmetry about the center of the beam span of the facing (x=0); the third and the fourth of Equation 21 represent the conditions at the free end of the facing (x=L); and the last two of Equation 21 represent the conditions of horizontal and vertical equilibrium, respectively, for the loaded facing of the considered sandwich beam, where $p_x(x)=p_z(x)=0$ (see Figure 3).

The last of the seven boundary conditions is derived by combination of the second of Equations 9 and the third of Equation 11. The resulting equation can be written in the form (Q is eliminated):

$$\frac{d^3q_z}{dx^3} + \frac{2\,t\,\alpha_1}{3\,\alpha_2}\frac{d^2q_x}{dx^2} + \frac{t\,\alpha_1}{6}\,q_x = \frac{t\,\alpha_1}{6}\,P_x\,. \tag{22}$$

The last boundary condition, supplementing the already established boundary conditions for the considered problem, is then specified by requiring fulfillment of Equation 22 at some position, say x=0 (center of beam span).

Thus, the complete set of boundary conditions necessary in order to determine the seven integrational constants has been completed, and the solution of the problem of the face-beam elastically supported by the core material has been achieved for the case of a sandwich beam in 3-point bending (Figure 3).

3 Superposition with Classical Sandwich Theory Solution

In order to obtain a complete solution to the specific problem of a sandwich beam in 3-point bending, it is necessary to investigate the overall bending and shearing effects; i.e., to establish an overall solution supplementing the local bending solution derived in the preceding sections of the present paper.

The simplest way to establish such an overall solution is by application of the classical theory of sandwich beams as given by Allen [1] or Plantema [2]. The present approach is based on the sandwich beam theory presented by Reference 1, to which the reader is referred for further details.

The transverse deflection of the considered sandwich beam can be written as:

$$w(x) = w_b(x) + w_S(x) \; ; \tag{23}$$

where $w_b(x)$ is the deflection induced by the overall bending, and $w_S(x)$ is the displacement induced by the overall shearing of the sandwich beam.

The displacement $w_b(x)$ is calculated by means of the ordinary theory of bending of beams, and for the considered problem (see Figure 3) the following is obtained:

$$w_b(x) = -\frac{P}{12\,D}\,(x^3 - 3\,L\,x^2 + 2\,L^3),\; 0 \le x \le L \; ; \tag{24}$$

where D is the flexural rigidity of the sandwich beam. D is given by (for the present case: 2b=1.0 mm):

$$D = E_f\,(2b)\left(\frac{t^3}{6} + \frac{t\,(2c+t)^2}{2}\right) + E_c\,(2b)\,\frac{2\,c^3}{3} \; . \tag{25}$$

The displacement $w_S(x)$ is expressed as:

$$w_s(x) = -\frac{P}{2\,A\,G_c}\,(L - x)\;,\; 0 \le x \le L : \tag{26}$$

where G_c is the shear modulus of the foam-core, and A is expressed by (2b=1.0 mm):

$$A = \frac{2b\,(2c+t)^2}{2c} \,. \tag{27}$$

The term AG_c, appearing in Equation 26, is referred to as the shearing stiffness of the sandwich beam.

The stresses (overall solution) in the faces and the core of the considered unit width sandwich beam (Figure 3) are expressed in the form; $0 \le x \le L$:

$$\begin{aligned}
(\sigma_x(x,z))_f &= -\frac{P\,(L-x)\,z}{2D}\,E_f, && \begin{pmatrix} (c+t) \ge z \ge c \\ -c \ge z \ge -(c+t) \end{pmatrix}; \\
(\sigma_x(x,z))_c &= \frac{P\,(L-x)\,z}{2D}\,E_c\,, && c \ge z \ge -c\,; \\
(\tau_{zx}(x,z))_c &= \frac{P}{2D}\left(E_f\,\frac{t\,(2c+t)}{2} + \frac{E_c}{2}\,(c^2 - z^2)\right), && c \ge z \ge -c\,.
\end{aligned} \tag{28}$$

The overall solution (Equations 23-28) can be superposed to the approximate local bending solution derived earlier. Thus, the total transverse displacement of the loaded facing due to overall bending, shearing, as well as local bending can be written:

$$w^{total}(x) = w_b(x) + w_s(x) + w_{local}\,(x)\,. \tag{29}$$

The local transverse displacement w_{local} is defined by the latter of Equation 1. The local displacement component w_{local} only defines the displacement of the "elastic line" of the loaded facing; i.e. the superposition of the overall and local deflection components suggested by Equation 29 will only describe the local bending effects at the interface between the loaded facing and core. It is not possible to decide, in explicit terms, the decay through the thickness of the core material of the deflections induced by local bending.

The stress field induced by the overall bending and shearing as well as local bending of the sandwich beam is another subject of major interest. The normal stress distribution in the loaded face-beam can be written in the form:

$$[(\sigma_x)_f]^{total} = [(\sigma_x)_f]_{overall} + [\sigma_x)_f]_{local}\,; \tag{30}$$

where $[(\sigma_x)_f]_{local}$ for a unit width beam can be expressed in terms of $q_z(x)$; $0 \le x \le L$

$$[(\sigma_x(x,z_{local}))_f]_{local} = \frac{M_{local} z_{local}}{I} \Rightarrow$$

$$[(\sigma_x(x,z_{local}))_f]_{local} = -\frac{E_f\, z_{local}}{K_z} \frac{d^2 q_z(x)}{dx^2}\ ;\ \frac{t}{2} \geq z_{local} \geq -\frac{t}{2}\ . \tag{31}$$

The coordinate z_{local} referred to in Equation 32 corresponds to the local face-beam coordinate shown in Figure 1.

The transverse normal stress component σ_z at the interface between the loaded face and the core is given by dividing the elastic response function $q_z(x)$ with the width of the beam (unit width), i.e., the interface transverse normal stress component can be written as:

$$[\sigma_z(x,z=c)]^{total} = \frac{q_z(x)}{2b}\ ;\ (2b = 1\ mm)\ . \tag{32}$$

Similarly the shear stress component at the interface between the loaded face and the core can be expressed as:

$$[\tau_{zx}(x,z=c)]^{total} = [\tau_{zx}(x,z=c)]_{overall} + \frac{q_x(x)}{2b}\ (2b = 1\ mm)\ . \tag{33}$$

As mentioned earlier for the total transverse displacements, the superposition of the overall and local stress field components only describes the state of stress in the loaded face and in the interface between the loaded face and the core of the considered sandwich beam. This area however, is by far the most interesting part of the structure, as it is here the peak values of the stress components will appear.

Thus, a complete (approximate) solution to the problem of a sandwich beam loaded in 3-point bending (Figure 3) has been derived by superposition of the local and overall displacement and stress fields respectively.

4 Results and Discussion

The applicability of the developed solution procedure will be demonstrated by an example, and the results obtained will be compared to a finite element solution in order to ascertain the quality of the suggested approach.

The example chosen in the present paper is the earlier described unit width sandwich beam in 3-point bending shown in Figure 3. The geometry, the material properties and the external point load P are as follows:

GEOMETRY: L=40.0 mm; t=1.0 mm; c=2.5 mm; 2b=1.0 mm;
FACE-BEAMS: E_f=15.0 GPa ("E-glass/epoxy");
FOAM-CORE: E_c=0.1 GPa; ν_c=0.35 ("PVC-foam": ρ_c=100.0 kg/m^3);
POINT LOAD: P=10.0 N/mm (force per unit width).

For the results presented in the following parts of the paper, the stress quantities will be normalized with respect to the "average" shear stress of the considered sandwich beam, which can be calculated by the expression [1]:

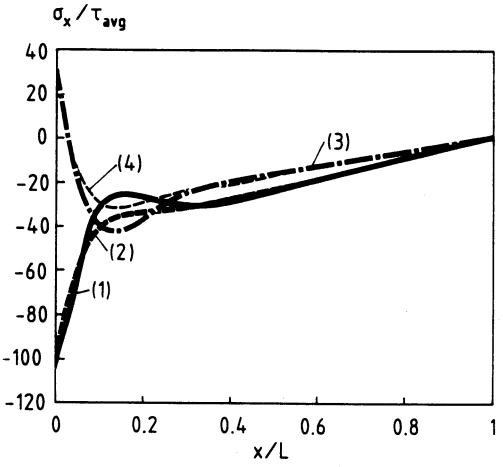
$$\tau_{avg} = \frac{P}{2\,(2c + t)} \,. \tag{34}$$

4.1 Comparison with Finite Element Solution

The software used for the model generation was the general finite element (FEM) program ANSYS, version 4.4A, developed and distributed by Swanson Analysis Systems, Inc. The sandwich beam shown in Figure 3, with geometrical and material properties as quoted above, was modeled using 2-D isoparametric 4-node plane stress/strain elements (element type referred to as STIF42 in the ANSYS 4.4A code) for the upper and lower facings as well as the core material. The upper and lower facings were modeled with two layers of elements, and the core was modeled using ten layers of elements. Due to the symmetry of the problem, only one half of the sandwich beam was generated. The total number of elements in the half-beam model was 1120 [9].

Figure 4 shows the distribution of normalized normal stresses σ_x/τ_{avg} along the upper and lower boundaries of the loaded facing of the sandwich beam. The results observed from Figure 4 are that the local bending effects are of significant importance near the point of external load application (x=0). At the upper surface of the facing a compressive state of stress is present (curves (1) and (2) in Figure 4) and the peak compressive value of σ_z/τ_{avg} is about -105. It is observed that the results obtained by the analytical and the FEM solutions, respectively, are very close to each other.

At the lower boundary of the facing a tensile state of stress is present at x=0, and this phenomenon is explained by the significant local bending contribution. Again a very close resemblance between the two solutions is observed. As the x-coordinate is increased, the local bending contribution is seen to fade out, and from x/L=0.3-0.4 the stress state corresponds to the linear variation of the overall bending moment which is a characteristic feature of the classical beam theory solution for the sandwich beam in 3-point bending.

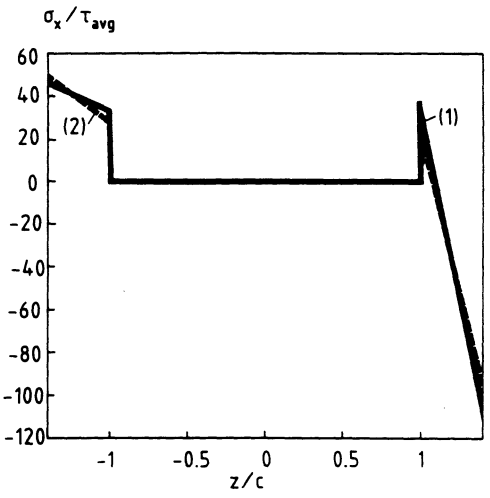


Upper Fibre: 1) ANALYT; 2) FEM ; Lower Fibre: 3) ANALYT; 4) FEM.

Figure 4. Longitudinal distribution of σ_x/τ_{avg} in upper and lower fibre of the loaded facing obtained by the analytical and FEM solutions.

Figure 5 shows the distribution through the thickness at x=0 of the sandwich beam of the normalized stress component σ_x/τ_{avg}. Curve (1) shows the results obtained by the analytical solution, and curve (2) shows the corresponding FEM results. The overall tendency is again a very close match of the results: the upper and lower facings of the sandwich beam are transferring the overall bending load and significant stresses are present. The stress state in the upper facing is determined by the overall bending together with the local bending contribution: at x=0 a compressive state of stress is present at the upper boundary, and at the lower boundary a tensile stress state is present. The stress state of the lower facing at x=0 is exclusively tensile since its is determined by the overall bending alone.

At this point, it should be mentioned that the very close match between the two solutions for the lower facing, shown in Figures 4 and 5, is a special feature of the example considered in the present paper, i.e., in general it cannot be expected that the classical sandwich theory solution will give accurate results for the lower facing. This circumstance can be explained by the fact (experimental results; [12]) that the decay of the transverse normal stress σ_z down through the core-material may not be complete, which again implies that transverse normal stresses have to be transferred between the core-material and the lower facing. As mentioned earlier, however, the question mark posed on the accuracy of the results obtained for the lower facing is no serious drawback for the suggested approximate solution procedure, as the loaded facing (where reliable results are obtained) will always be the most interesting part of the sandwich structure (most severely loaded).



1) $(\sigma_x/\tau_{avg})_{ANALYT}$; 2) $(\sigma_x/\tau_{avg})_{FEM}$.

Figure 5. Transverse distribution at x=0 of σ_x/τ_{avg} obtained by the analytical and FEM solutions, respectively,

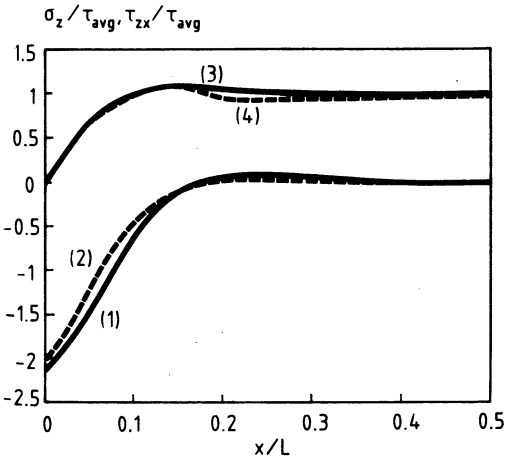
Figure 6 shows the distribution of the normalized transverse normal and shear stress components, σ_z/τ_{avg}, τ_{zx}/τ_{avg} at the interface between the loaded face-beam and the core material. Focusing at first on the curves (1) and (2) of Figure 6, showing the distribution of σ_z/τ_{avg}, it is observed that the results obtained by the analytical and FEM solutions, respectively, show a very close match. The predicted peak values (x=0) are very close to each other, and the characteristic decay with increasing values of x is also observed for both solutions. Furthermore, the wavy harmonic character of the "decay function" predicted by the elastic foundation formulation is confirmed by the finite element solution.

Curves (3) and (4) of Figure 6 show the interfacial distribution of τ_{zx}/τ_{avg} obtained by the two solution methods, and again a very close resemblance of the results is observed.

Other results regarding the comparison between the developed approximate solution procedure and the FEM model of the sandwich beam could be shown, but they all show the same close match as observed from Figures 4, 5 and 6, and no further comparative results will be given in the present paper (for further information see [9]).

4.2 Parametric Study

In order to illustrate the influence of certain characteristic parameters on the local interfacial stress distribution in the near vicinity of the point of external load application, the results obtained from a brief parametric study will be presented. The characteristic material and



1) $(\sigma_z/\tau_{avg})_{ANALYT}$; 2) $(\sigma_z/\tau_{avg})_{FEM}$; 3) $(\tau_{zx}/\tau_{avg})_{ANALYT}$; 4) $(\tau_{zx}/\tau_{avg})_{FEM}$.

Figure 6. Distribution of σ_z/τ_{avg}, τ_{zx}/τ_{avg} at the interface (z=c) between the loaded facing and the core material.

geometrical parameters included in the analysis are: the ratio between the elastic moduli E_f/E_c, and the thickness t of the loaded facing.

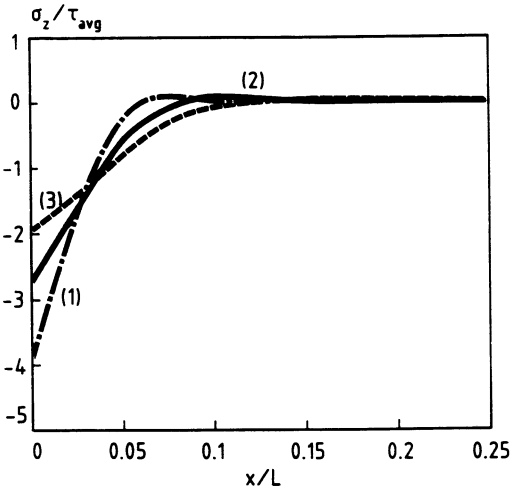
The parameters L and c (see Figure 3) also influence the stress distribution within the sandwich beam, but they only exert influence on the overall bending and shearing of the sandwich beam, and therefore they will not be included in the presented parametric study.

Figures 7 and 8 show the effects of altering the stiffness ratio E_f/E_c on the interfacial distribution of the normalized transverse normal stress, σ_z/τ_{avg}, and the normalized local shear stress, τ_{zx}/τ_{avg}. The thickness of the loaded face is chosen to t=1.0 mm, and E_f/E_c is given the values 50, 150 and 450.

From Figure 7 it is seen that the lower the value of the stiffness ratio, E_f/E_c, the higher the peak value of σ_z/τ_{avg}. Furthermore, it is observed that the wave-length of the elastic response is increased significantly as E_f/E_c is increased.

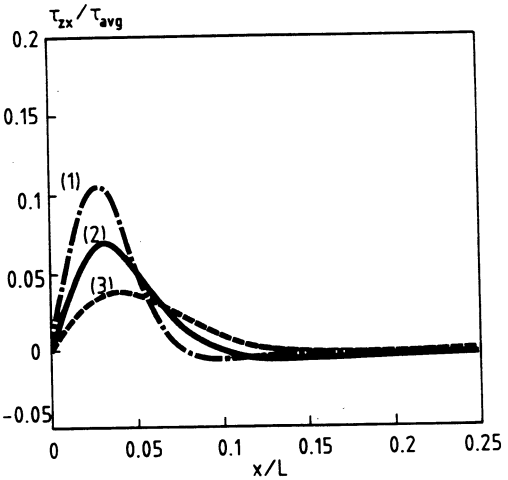
Similar results are observed from Figure 8, except that the peak value of τ_{zx}/τ_{avg} (representing only the shear contribution induced by local bending) is located some distance away from x=0. Again it is observed that the wave-length of the elastic shearing response increases with increasing values of E_f/E_c.

The results shown in Figure 8 also indicates when it is advisable to use the two-parameter elastic foundation model, suggested in the present approach, instead of the simpler Winkler foundation model (one-parameter elastic foundation model) used by Weissman-Berman et al. [6]. It is observed that the peak value of τ_{zx}/τ_{avg} decreases very rapidly as E_f/E_c is increased, and for large values of the stiffness ratio the build up of interfacial shear stresses due to local bending is negligible in comparison with the other stress components. In this case the results



1) E_f/E_c=50.0; 2) E_f/E_c-150.0 ; 3) E_f/E_c=450.0

Figure 7. σ_z/τ_{avg}. vs. x/L at the interface between the loaded facing and the core material (E_c fixed to E_c=0.1 GPa).



1) E_f/E_c=50.0; 2) E_f/E_c-150.0 ; 3) E_f/E_c=450.0

Figure 8. τ_{zx}/τ_{avg} vs. x/L at the interface between the loaded facing and the core material (E_c fixed to E_c=0.1 GPa).

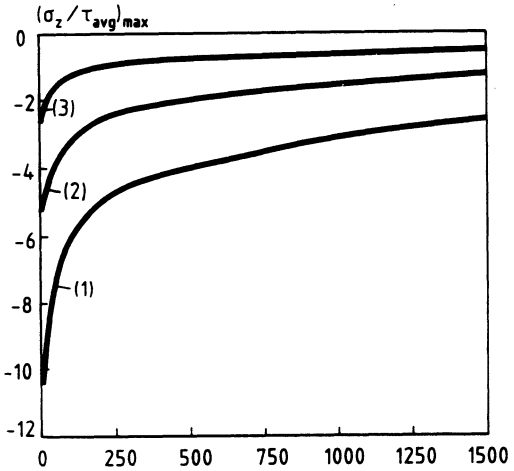
obtained by the two-parameter elastic foundation model will be nearly identical to the results predicted by use of the Winkler foundation model (further treatment in Section 4.3).

Curves similar to the ones shown in Figures 7 and 8 can be obtained by altering the thickness t of the loaded facing for a fixed value of E_f/E_c, but they will not be shown in the present paper, since the conclusions drawn are similar to the conclusions drawn from Figures 7 and 8. The overall tendencies are that the peak values of σ_z/τ_{avg}, τ_{zx}/τ_{avg} decreases

significantly as t is increased, i.e., as the flexural rigidity of the face-beam is increased. Furthermore, the wave-length of the elastic response functions increases with increasing t-values.

In order to present the main results, obtained from the parametric studies quoted above, in a more compressed and informative form, a few plots showing the peak values of σ_z/τ_{avg}, $\tau_{zx}^{total}/\tau_{avg}$ as functions of the stiffness ratio E_f/E_c as well as the thickness t have been prepared. The superscript "total" refers to the shear stress components obtained by superposition of the local and overall shearing contributions.

Figure 9 shows the peak value (x=0) of the interfacial transverse normal stress component $(\sigma_z/\tau_{avg})_{max}$ as function of E_f/E_c. Three different curves are shown, each representing different values of the face-beam thickness t (t=0.5, 1.0 and 2.5 mm).



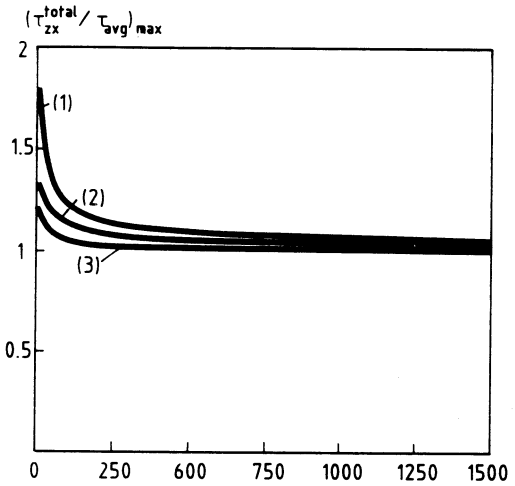
1) t=0.5 mm ; 2) t=1.0 mm ; 3) t=2.5 mm .

Figure 9. $(\sigma_z/\tau_{avg})_{max}$ vs. E_fE_c.

It is seen that $(\sigma_z/\tau_{avg})_{max}$ shows a strong dependency of E_f/E_c as expected from the results shown in Figure 7. A significant feature of the plots shown is that the peak value of σ_z/τ_{avg} attains very large values for very small values of E_f/E_c, even though it should be recalled that the elastic foundation approach, on which the results are based, becomes inadequate for very small values of E_f/E_c (corresponding to deflections of short wave-length).

It is also observed that the peak value of σ_z/τ_{avg} is influenced by the face-beam thickness t in the way that the larger the value of t (i.e. the larger the flexural rigidity of the face-beam) the lower the peak values of σ_z/τ_{avg}.

Figure 10 shows the peak values of the interfacial shear stress τ_{zx}/τ_{avg} as function of the stiffness ratio E_f/E_c for three different values of the thickness t of the loaded facing (t=0.5, 1.0 and 2.5 mm). It is recognized that the peak values of τ_{zx}/τ_{avg} are strongly dependent on the E_f/E_c-values in the way, that very large peak shear stresses are obtained for very small values of the stiffness ratio (it is emphasized that the calculated peak stress values for low values of E_f/E_c are questionable as mentioned earlier), and that the peak values of $\tau_{zx}^{total}/\tau_{avg}$ decreases strongly as the stiffness ratio is increased. The peak value of $\tau_{zx}^{total}/\tau_{avg}$ approaches unity asymptotically as E_f/E_c goes to infinity, i.e., the interfacial shear stress contribution induced by local bending diminishes as the stiffness ratio becomes large.



1) t=0.5 mm ; 2) t=1.0 mm ; 3) t=2.5 mm .

Figure 10. $(\tau_{zx}/\tau_{avg})_{max}$ vs. E_f/E_c.

Again it is clear, that the peak value of $\tau_{zx}^{total}/\tau_{avg}$ depends on the face thickness t in the way, that the larger the value of t, the lower the peak value of $\tau_{zx}^{total}/\tau_{avg}$.

4.3 Discussion of Range of Applicability

The introduction of the elastic foundation analogy, in the context of local bending of sandwich panels subjected to localized loads, has been carried out in order to present a simple method of obtaining detailed information about the displacement and stress fields induced locally (loaded facing and interface between loaded facing and core material) as a result of localized lateral loads.

The basic idea behind the application of an elastic foundation analogy is, as described earlier, to approximate the supporting medium (core material of sandwich beam) by continuously distributed linear tension/compression and shear springs (two-parameter foundation model). The quality of the approximation, however, is strongly dependent on the quality of the basic assumptions of the elastic foundation model employed.

The basic assumptions of the two-parameter elastic foundation model are given by Equation 1, which states that the elastic response of the supporting medium at a given position (specified by the longitudinal coordinate x; see Figure 1) is directly proportional to the displacements of the lower boundary of the loaded facing.

This implies, that the foundation moduli K_z and K_x are assumed to be constants, which can be related to the elastic coefficients of the core material as well as the elastic coefficients and the geometrical characteristics of the loaded facing.

Whether assumptions of the type described are generally justified for typical core materials (polymeric foam cores, honeycomb cores, balsa cores) used for structural sandwich structures is a matter of discussion, and the most important questions posed are:

1. Is the assumption of "constant-value" foundation moduli K_z, K_x justified?
2. Is the assumption of linear elastic material properties realistic?
3. Can the elastic foundation analogy be used to model all types of core materials typically in use for structural sandwich constructions?
4. What is the difference between the Winkler and two-parameter foundation models from a practical engineering point of view, i.e., when is it recommended to use one model instead of the other?

Concerning the first of the questions posed, the answer demands some additional considerations. The assumption of constant value elastic foundation moduli K_x, K_z, is obviously not generally justified, but can, at the most, be legalized if some, as yet unspecified, further restrictions are imposed on the class of problems for which the proposed elastic foundation analogy can be used successfully.

The considerations leading to these further restrictions, on the range of applicability of the elastic foundation analogy, will not be given in detail in the present paper, but the reader is instead referred to the analysis presented in Reference 1 (pp. 169-171) concerning the bounds of application on the Winkler hypothesis in the context of analysis of wrinkling (local

instability) of structural sandwich panels. Furthermore, the reader is referred to Reference 9 (pp. 21-24) in which the bounds on the two-parameter elastic foundation model are discussed.

It is shown in References 1 and 9, that the key to the specification of the necessary restrictions on the applicability of elastic foundation models (as the ones discussed) is the wave-length (denoted by λ) of the deflections of the elastically supported loaded facing. The conclusion drawn from the reflections presented in the quoted references is that the elastic foundation models (Winkler as well as two-parameter foundation model) become inadequate for deformations of short wave-length. This is caused by the fact that the shearing deformations of the supporting medium becomes increasingly important as the wave-length of the deflections decreases. Thus, it is recognized that the simplifications introduced by the application of an elastic foundation model, instead of a continuum model for the core material, are not justified for deformations of short wave-length.

A natural question in this context is, whether it is possible to give an explicit quantification of the concept of "deformations of short wave-length". Unfortunately the answer to this question turns out to be negative if the term "quantification" means definition of very precise bounds on the applicability of the foundation models. This is caused by the circumstance that the wave-length of the deflections is related to the characteristic material and geometrical parameters of the problem in a very complex manner.

For practical sandwich panels, however, the bounds imposed by the vaguely formulated concept of "deformations of short wave-length" are not likely to be active, since the typical face thicknesses (0.5 mm<t<10.0 mm), and the typical stiffness ratios ($25<E_f/E_c<1500$), will usually ascertain sufficiently large deflectional wave-lengths to ensure the justification of the simple elastic foundation approach instead of a very complex two or three dimensional continuum formulation.

The second question posed was whether it is reasonable to assume linear elastic properties of the constituent materials of typical structural sandwich panels. The answer to this question is not a very direct and simple one, as some face materials as well as some core materials can be said to behave reasonably linear elastically, and some do not.

However, the service conditions, under which structural sandwich panels are employed, are usually sought to be well within the safe domain specified by the proportional limit of the constituent materials. Thus, it is expected that the suggested approach, based on linear elastic assumptions, is capable of giving a fairly good estimate of the magnitude of the stress concentrations induced by local bending effects.

The third question posed was whether the elastic foundation approach can be used to model all types of core materials in use for structural sandwich panels, and in order to give a proper answer to this question it is necessary to include some additional comments.

There are three major types of core materials used for structural sandwich panels: 1. polymeric foam core materials, 2. honeycomb core materials, and 3. balsa cores. The first and third of the listed core material types can be considered as being homogeneous materials (at least from a macroscopic point of view which is usually preferred in the context of engineering applications), since the cellular structure of these materials is usually at least one order of magnitude smaller than the characteristic dimensions of the considered sandwich panel (face thickness, core thickness, width of sandwich panel, sandwich panel span width etc.). Thus, it seems reasonable to model the foam and balsa core materials respectively by using a continuum or an elastic foundation formulation as suggested.

With respect to the honeycomb type cores, it is a bit more complicated to present an answer to the posed question. Honeycomb cores are discrete by nature, i.e., they do not support the facings of a sandwich panel continuously, but rather in a discrete manner along the edges of the honeycomb cells. This means, that continuum formulations or continuous spring support formulations (Winkler and two-parameter models) may be incapable of modelling the core material realistically. Whether this is the case, is determined from the cell size of the honeycomb cell structure: if the characteristic dimensions (width and edge length of cells) of the cells are small compared to the thickness of the facings, continuum or continuous spring support formulations will be capable of giving a fair estimate of the mechanical behavior from an engineering point of view. If, on the other hand, the cell size of the honeycomb structure is comparable to, or even larger than, the face thicknesses, it is not likely that good results can be obtained, as the facings will tend to act like plates within the boundaries of each cell in the honeycomb structure.

The above mentioned considerations deals with the justification of elastic foundation models in general as opposed to elastic continuum models, with a discussion about the assumption of linear elastic material properties, and with a discussion about the capability of continuum or elastic foundation formulations to describe the mechanical behavior of various types of core materials. No distinction between the very simple Winkler foundation model and the more complicated two-parameter model has been presented.

It is clear that none of the foundation models will give satisfactory answers for "deformations of short wave-length", as it can be shown [1,9], that it is not possible to select constant values of the foundation moduli K_z and K_x which are appropriate for displacements of any wave-length. The reason for this is, as described earlier in this section, the fact that the shearing deformations of the core material becomes very influential for deformations of short wave-length.

This last statement, however, makes it possible to distinguish between the "quality" of the Winkler and two-parameter foundation models, because the latter actually does take into account the possible shearing interaction (although in very simple form) between the loaded facing and the core material of a sandwich beam subjected to strongly localized lateral loads.

Thus, it is recognized that the two-parameter foundation model is superior to the Winkler foundation model, as it predicts the existence of the interfacial shear stress distribution function $q_x(x)$, although the formulation of the shearing interaction effects is not sophisticated enough to handle problems characterized by displacements of very short wave-length.

The application of the two-parameter foundation model instead of the simpler, from a mathematical point of view, Winkler foundation model can be recommended for structural sandwich panels characterized by the following approximate relationship:

$$\lambda \approx 5.03\ t \sqrt[3]{\frac{E_f}{E_c}} < 50\text{–}60\ [\text{mm}]\ ; \tag{35}$$

where λ is the wave-length of the elastic deformations (elastic line). The guidelines defined by the inequality (35) should only be considered as a rough estimate of the recommended bounds of applicability of the two foundation models under consideration. Thus, it is possible to imagine structural sandwich panels characterized by extreme values of the thickness t or the stiffness ratio E_f/E_c giving rise to short wave-length elastic responses, even though λ lies well within the "safe domain" specified by Equation 35, i.e., in the domain where the Winkler hypothesis should be able to supply sufficiently accurate results.

The values of λ suggested by the inequality (35) as an appropriate applicational separation between the Winkler and two-parameter foundation models, however, will usually ensure that the peak value of the interface shear stress component $(\tau_{zx})_{local}$ is negligible compared to the interface peel stress component $(\sigma_z)_{local}$.

5 Concluding Remarks

A method of analyzing the local stress and displacement fields in the near vicinity of strongly localized external loads applied to sandwich beams (or plates in cylindrical bending) has been presented. The developed formulation is based on the assumption that the deflection of the loaded facing against the not loaded facing of a laterally loaded sandwich beam can be properly modeled by use of an elastic two-parameter foundation model. Furthermore, it is assumed that the facings as well as the core material behave linear elastic.

The local stress and displacement fields derived by application of the two-parameter elastic foundation model are superimposed to the stress and displacement fields derived by use of the classical theory of sandwich beams (or a rough FEM model), and an overall solution is obtained including overall bending and shearing as well as local bending effects.

The applicability of the derived solution procedure has been investigated by analyzing a test-example: a sandwich beam in 3-point bending. The results obtained by use of the

developed method have been compared with a finite element solution, and an excellent match of the computed solutions has been observed.

Thus, it is has been shown that it is possible to estimate (with a high degree of accuracy) the severity of the stress concentrations in the loaded facing and in the interface between the loaded facing and the core of sandwich beams by application of a suitable elastic foundation formulation.

A brief parametric study, based on the developed solution procedure, has shown that the local bending effects are very strongly influenced by the stiffness ratio E_f/E_c and the thickness t of the loaded facing. The peak values of the interfacial transverse normal and shear stresses attain very large values for small values of both E_f/E_c and t, corresponding to elastic responses of short wave-lengths. Furthermore, it is observed that these peak values of the interface stresses decreases rapidly as E_f/E_c and t are increased.

From the results obtained in the parametric study, it is recognized that the only way to reduce the locally induced stress concentrations is to increase either of E_f/E_c or t. From a practical point of view, however, it is not possible to change E_f/E_c locally. Thus, it is seen that t is the only parameter which (realistically) can be adjusted locally in order to accomplish a less severe distribution of interface stresses.

On the basis of the elastic foundation analogy presented in the present paper, graphical "design-charts" have been prepared for different load cases (point load, uniformly distributed load, concentrated bending moment [11]).

For a specific load case it is possible, from these graphical "design-charts" (similar to Figures 9 and 10) to find peak values of the interface stress components, as well as the peak value of the bending stress in the loaded facing, for specified values of the stiffness ratio E_f/E_c and the face-thickness t. These peak stresses, which are induced by local bending effects, can be superposed to the stresses obtained by an overall solution, i.e., a classical sandwich theory solution or a rough finite element solution obtained by use of sandwich beam or plate elements.

Thus, graphical "design-charts" [11] can be used in a design process, as a very cost effective Concurrent Engineering tool, to give a fairly accurate estimate of the stress concentrations induced by local bending, and among the type of questions which can be answered are:

1. Considering a sandwich with given face and core materials, given face- and core-thicknesses; what magnitude of external load could be applied?
2. Considering a sandwich with given face material and face-thickness, given core material with given compressive strength; what face-thickness is required to prevent core-failure by crushing?

3. Considering a sandwich with given face and core materials, given face- and core-thicknesses, given total external load; over how large an area should the external load be distributed to prevent the maximum allowable stresses (core and facings) to be exceeded?

The approximate solution procedure presented is only strictly valid for sandwich beams, but it is easily extendable to the case of sandwich plates in cylindrical bending. Plate analysis in general, however, cannot be accomplished directly with the solutions presented so far, but general sandwich plate solutions, based on an extended version of the two-parameter foundation model, will be presented in the near future.

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Symbols Used

Latin Symbols

A	:	$b(2c+t)^2/c$ (mm^3);
A_j	:	(j=0,...,6) integrational constants;
b	:	half width of sandwich beam (mm);
c	:	half thickness of core material (mm)
D	:	flexural rigidity of sandwich beam (Nmm^2)
E_c,E_f	:	elastic moduli of core and face materials respectively (GPa);
G_c	:	shear modulus of core material (GPa);
I	:	area-moment of inertia of facing (mm^4);
K_x,K_z	:	elastic foundation moduli (GPa);
L	:	half length of sandwich beam (mm);
M	:	bending moment (Nmm);
N	:	normal force (N);
p_x,p_z	:	external surface loads (N/mm);
P	:	external point load measured per unit width (N/mm);
q_x,q_z	:	interfacial stress distribution functions (N/mm);
Q	:	shear force (N);
t	:	thickness of faces (mm);
u	:	longitudinal displacement (mm);
w	:	lateral displacement (mm);
x	:	longitudinal coordinate (mm);
z	:	lateral coordinate (mm).

Greek Symbols

α_j	:	(j=0,1,2) coefficients;
ε_x	:	longitudinal normal strain;
η	:	coefficient (root to characteristic equation);
λ	:	wave-length of elastic response (mm);
ν_c	:	Poisson's ratio of core material;
ξ	:	coefficient (root to characteristic equation);
ρ_c	:	density of core material (kg/m^3);
σ_x,σ_z	:	normal stress components (MPa);
t_{zx}	:	shear stress component (MPa);
ϕ_1	:	coefficient (root to characteristic equation).

Superscripts

total	:	superposition of local and overall bending contributions.

Subscripts

avg	:	average;
b	:	bending;
c	:	core;
f	:	facing;
local	:	contribution induced by local bending;
0	:	neutral axis (loaded face-beam) quantity;
overall	:	contribution induced by overall bending;
s	:	shear.

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